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Technical and Business Writing (SS2012)

Date: 17th December, 2024

Course Instructor(s)

Mr. Muhammad Zulfiqar Ali

Final Exam

Total Time (Hrs.):	**3**
Total Marks:	**100**
Total Questions:	**4**

Attempt all the questions in the answer copies.

________	**BEE 7A**	________
Roll No	Section	Student Signature

Question 1: Part A. *{You should spend about 25 minutes on this task.}* **[CLO – 2] (25)**

Case Scenario: As an electrical engineer working for a company that designs and manufactures electrical systems for residential buildings, you are tasked by your Project Manager, Mr. Zafar, with writing an informal technical report (use memo format) on a new energy-efficient lighting system that your team has developed. The report should explain the key technical features, advantages, and expected performance of the system. However, during the development phase, there were some challenges related to product testing, which led to delays in the project timeline. You must assume any data and present in tabular format and support through infographics with figure labels to validate your report.

In your report you should:

1. **Describe** the technical features of the lighting system, explaining how it improves energy efficiency compared to traditional systems. (Knowledge/Comprehension)
2. **Analyze** the ethical considerations that must be addressed when communicating the delays in the testing phase to stakeholders, ensuring transparency and honesty in your reporting. (Analysis)
3. **Justify** the ethical principles you will follow to maintain the credibility and trust of your audience while addressing the testing delays in the final report. (Evaluation)
4. **Propose** how you would use ethical communication strategies to effectively present the situation and ensure that stakeholders understand the reasons behind the delays and the steps taken to resolve them. (Synthesis) (15)

Question 1: Part B: Case study – the ethical dilemma. *{Needs about 30 minutes.}* [CLO, 2] (10)

Personalize that you are collaboratively working with a team of technical writers at New-Line-Energies Pvt. Ltd. They are producing a collaborative document for the clients on how to use the company's solar appliances and other electronic products. Per your part, you have been asked to create the user manual for a dual mode solar powered automatic gas oven. The Head of the marketing team, however, approaches you with a request to highlight certain features of the product, somehow exaggerating its functional utility. He argues that this would make the product more appealing to potential customers, leading to increased sales and market share. They want you to use language that implies that their automatic solar-powered oven surpasses all the competitive brands, even if the data supporting this claim is not conclusive.

Your supervisor, who is also aware of the situation, leaves the decision up to you, but also hints that you do meet the expectations for the company's sales and profits.

1. What ethical dilemma is presented in this scenario? Explain the conflicting interests and values that occur in this context. (3)
2. Elaborate on at least three ethical considerations that you, as a technical writer, need to consider when deciding how to handle the situation. (3)
3. Evaluate the potential consequences of both accepting and rejecting the marketing team's requests on the company and the customers. (2)
4. Based on your ethical analysis, propose three strategies *(Only list the points)* that you believe align with the best ethical practices for technical writing. (2)

Question 2: Part A. *{You should spend about 25 minutes on this task.}* **[CLO, 3] (25)**

(a). Read the following sections of a research article and **formulate an Abstract** based on the information you acquired from it. Develop one cohesive paragraph. You should mention the name of each component of the Abstract in brackets. Finally, write four key words below the Abstract. (8+2=10)

'The number of students enrolled in tertiary education today is higher than in previous decades because of increased family revenues. At the same time, competition on the labor market is increasing worldwide. According to the Euromonitor education trends published in 2018, there is also a constant rising of the percentage of female students, from 50% in 2011 to 51% in 2016 [1]. Employability levels for women, however, do not grow at the same rate. At the same time the recruitment pool for students remains stable, in spite of declining fertility rates and an ageing population, and enrollment in higher education increased. Students are also more mobile internationally, seeking quality education and being eager to acquire more technical skills, which represent a strength in getting a new or a better job. In this context, the responsibility of universities towards their students increased. Several important changes were conducted in educational management worldwide, but there is still room for improvement. Universities must ensure their graduates acquire relevant skills for the labor market for a long period of time. The information technology sector is expected to continue to grow in the coming years and this will affect all industry sectors. Given these, the objective of this study is to investigate the employability of Romanian students who graduated from the Faculty of Electrical Engineering at the Technical University of Cluj-Napoca. A special focus will be given to the extent to which these graduates choose entrepreneurship as a career path, in accordance with the meta competencies that students are expected to achieve nowadays. A short review of the existing trends regarding the employability in global, European and local electric field is provided. The reports of the skills required by the labor market are cited in order to compare the findings of the study we conducted. II. LITERATURE REVIEW A. Electric Engineering Employability in Romania It is well known that Electrical engineering is one of the most complex fields of engineering. It is a broad field that includes many subdomains. For example, electro mechanic specialization trains engineers to be capable of problem solving specific to technological development through the design, implementation and operation of electromechanical systems and equipment. [2] Electrical engineers are meant to design, develop, test, and maintain the electrical equipment, devices and systems which use electricity, electronics, and electromagnetism. The term "employability" can be interpreted in several ways, depending on the different needs of the labor market. In the frame of the Bologna process, employability refers to "the ability of one person to be accepted for employment in accordance with their competencies, to maintain them through professional development, to have entrepreneurial abilities and to be competitive on a long period of time in order to quickly adapt to the labor market". [3] Existing studies conducted in Romania suggest imperative requirements for improving students' skills in engineering, transportation, and distribution [4][5]. Although the percentage of graduates in science, technology, engineering, and mathematics (STEM) as a total of higher education graduates is among the highest in the EU, the number of graduates actually available for recruitment on the labor market is insufficient and constantly diminishing. Reports agree that graduates often have in depth theoretical knowledge, but sometimes practical or the soft skills are neglected in educational process [6]. In addition, in Romania, the employment rate of the universities' graduates reached a percentage of 89% in 2018, higher than EU average: 85.5%. At the same time, no formal procedures are out in place in Romania, regarding the collection of skills required/industry sector. [7] In 2015, the Romanian Court of Accounts conducted a study regarding the graduates' employment rates, six months after graduation. Findings reveal that the highest rates are registered among engineering graduates, with a percentage of 84%. In the same research, electrical engineering graduates had employment rates of up to 68% (employed exactly in their specialization). This percentage was lower than in 2006 (80%). However, the results also indicated that electrical engineering graduates had higher employability rates than those graduating from electro mechanics (50% in 2016, declining by 25% from 2006). [3] In Romania, fostering the entrepreneurs' mindset is only at the beginning, in spite of the fact that, to assess the entrepreneurial and innovative potential of universities, Romania participates in the HEInnovate program initiated by the OECD and the European Commission. The Ministry of National Education (MEN) in collaboration with an NGO - Junior Achievement Romania applies HEInnovate in Romanian universities, which consists in the process of diagnosis and evaluation of the current state of action along with entrepreneurial and innovative approaches in universities, using the HEInnovate tool. Another measure consists in organizing Innovation Days among students, together with the local community, in order to identify the potential solutions for the challenges resulting from the analysis. Furthermore, beginning with 2017, MEN supports universities to implement actions aimed at facilitating the activities of student entrepreneurial societies by introducing a new strategic area dedicated to the establishment and support of activities in the frame of student entrepreneurial societies. [16] The present study was conducted on a sample of 214 alumni from the following specializations: Electro mechanics (109 engineers) and Economical Engineering in Electric, Electronics and Energetic fields (105 engineers). The engineers in the sample had graduated from the Electrical Engineering Faculty from the Technical University of Cluj -Napoca in the period 2013-2019. A database comprising all positions students had occupied from graduation up to the date of the study was set up. Positions were split in more categories, based upon clusters techniques, being mentioned the employment periods of time, as well: • Type: Internship / Practice/Employee; • Specialization: Electro mechanic, Electrical Engineering, Sales/commercial Engineers, Financial, I.T., Others; • Specific of work: Research, development/design, execution/production; • Managerial positions: Entrepreneur, manager, nonmanager; • Period of time: under six month, 6 month-1 year, 1-2 years, 2-4 years, over 4 years. In the first step, all the alumni were assessed from secondary data sources, such LinkedIn or social media profile, while in the second step, we sent out an e-mail survey, aiming at collecting up to date data for every person in the sample. From our list, a percent of 10 percent of respondents were removed because data retrieved from secondary data sources was obsolete and researchers had been unable to contact the graduates directly. The results obtained reveal that 51% of former students started their careers in the field of execution, respectively 37.1% in the field of research &development at companies such as Bosch (Cluj), Tenaris (Zalau) or Porsche Engineering (Cluj), Emerson (Cluj), which provided them the opportunity to train themselves in the frame of many internships. According to the data analyzed, it is observed that over 32% of former graduates started their careers with payed internship programs or compulsory internships during their years of study. The secondary data along with questionnaire information was computed in order to describe Electrical Engineering from Technical University of Cluj-Napoca study case. In conclusion, the employment rate after graduating Electrical Engineering Faculty is high and in connection with the environmental factors such as the development of IT automotive and research-development market. The companies' requirements are founded at the conjunction between electric field, automotive and electronics. The Technical University graduates are mainly employed in their specialization, and a third of them were enrolled first in an internship program, fact that emphasize the importance of the practical side of educational process. The majority of graduates works in executive jobs and up to a third of them are involved in research -development activities. Regarding the entrepreneurial choice, there is a percentage of 2.8% entrepreneurs in the sample, slightly lower that the national rate of entrepreneurs who graduated technical higher education (3.8%). These findings could not demonstrate that current curricula sustain and catalyzes the students in becoming entrepreneurs, but the results can be connected with the individual mindset of the graduates.'

(b). Construct a catchy, relevant title to this research work and draw a rectangle around it. (5)

Question 2: Part B. Write an effective and cohesive paraphrase of the following passage using in-text IEEE citation sources given at the end. Avoid text-lifting. *{Utilize about 15 minutes}* **[CLO, 3]** (10)

The integration of renewable energy sources, such as solar and wind power, into existing electrical grids presents several challenges, including intermittency and variability of supply. Advanced energy storage systems, such as lithium-ion batteries, can mitigate these issues by storing excess energy during periods of high generation and releasing it when demand exceeds supply. Moreover, smart grid technologies enable more efficient management of electricity distribution, optimizing energy usage and reducing waste. As renewable energy adoption grows, these technologies are expected to play a pivotal role in enhancing grid reliability and promoting sustainable energy practices. Continued research and development are essential to improving the efficiency and cost-effectiveness of these technologies for widespread deployment in various regions.

IEEE Citations:

1. *J. Smith et al., "Advances in renewable energy integration," IEEE Trans. on Energy Conversion, vol. 32, no. 3, pp. 502-510, 2019.*
2. *A. K. Gupta, "Energy storage systems for grid stability," IEEE Power Energy Mag., vol. 18, no. 2, pp. 65-72, 2020.*
3. *M. Zhang and P. Liu, "Smart grids and their impact on electricity distribution," IEEE Electr. Power Syst. Research, vol. 58, no. 4, pp. 240-247, 2021.*

Question 3: Part A. *{You should spend about 15 minutes on this task.}* **[CLO, 4]** **(25)**

Construct an email to the Head of the Electrical Engineering Department, FAST-NUCES, Lahore Campus, Prof. Dr. Saima Zafar, requesting her to let you work on a collaborative project with a local technology company. The project aims to design a smart energy management system for commercial buildings (you may assign any name to the project). You should tell her why you deserve to be selected. Your email must have strong supportive evidence/ facts (assume any relevant details). <u>You must compose an effective Subject Line.</u> (10)

Question 3: Part B. *{You should spend about 30 minutes on this task.}* **[CLO, 4]** (15)

Assume that you are working in a company that provides problem-solving services related to industrial electrical and electronics issues. Angro-Agri Services have approached you to analyze and seek solutions to a major problem that occurred at one of their fertilizers producing units at Sheikhupura (you may assume any real-life engineering problem). As an electrical engineer, **Formulate** and **propose** a comprehensive, formal Problem Analysis and Solutions Report in a Memo Format (350-450 words) to your client. You may assume any relevant data to validate your work. Present this data in tabular and infographics. You must address the following:

- The purpose of writing
- Summary/ background of the problems
- Graphics/ tabular data/ stats
- Consequences of the problems
- Restatement of the problems
- Degree of urgency in handling them
- Recommended three solutions

Question 4. *{You'll need about 25 minutes for this answer}* **[CLO, 5]** **(25)**

'Personal Statement' forms a key part of your university application. The assertiveness of this application bears a high significance in your admission decisions and determines the direction of your future contributions. Assume you have completed your BS Electrical Engineering degree from FAST NU-Lahore, and you have done a one-year internship at an organization *(assume any from your choice field)*. Presently, you are highly motivated and aspire to continue higher education in a relevant field at the University of Toronto, Canada. You contacted the University for admissions, and they have asked, as a partial fulfillment of admission requirements for MS program, to write a Personal Statement in 300 – 350 words to showcase how your skills, experience, and aspirations make you a good fit for this program.

Your personal statement should precisely include the following:

- Your Introduction
- Professional experience
- Problem-solving and leadership skills
- Reasons for choosing the program
- Short term and life-long goals
- Expected contribution to the field
- Extracurricular activities

NOTE: Please write clearly and concisely. Avoid using filler words and multiple meaning ambiguous phrases. Do Not change the original numbers of the questions.